



# Zoo Network Mainnet Launch: Technical Checklist and Rehearsal Plan

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**Version 2026.04**

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April 2026

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# 1 Overview

This document is the authoritative checklist for ZOO Network mainnet launch. It covers pre-launch validation, rehearsal on staging, launch-day sequence, post-launch operations, and rollback procedures.

ZOO Network is a Layer 2 on LUX mainnet comprising 14 virtual machines (chains) and one Layer 3 subnet (Beluga). All chains share LUX consensus and settle to LUX P-Chain.

## Clusters:

- **lux-k8s** — LUX mainnet validators (DOKS, SFO3)
- **lux-test-k8s** — Testnet / staging (DOKS, SFO3)
- **hanzo-k8s** — Shared platform services (IAM, KMS, gateway)

**Status legend:** [ ] = pending, [x] = complete, [!] = blocked.

# 2 Pre-Launch: Testnet Validation

All items must be [x] before rehearsal begins.

## 2.1 VM Verification

| #  | Chain         | Criteria                                         | Owner  | Status |
|----|---------------|--------------------------------------------------|--------|--------|
| 1  | EVM (C-Chain) | Standard EVM conformance tests pass              | Core   | [ ]    |
| 2  | DEX (D-Chain) | Order matching, AMM, limit orders verified       | Core   | [ ]    |
| 3  | FHE (F-Chain) | Encrypted compute round-trip, TFHE gates correct | Crypto | [ ]    |
| 4  | I-Chain       | DID issuance, revocation, resolution             | Auth   | [ ]    |
| 5  | K-Chain       | Key management, TLS cert provisioning, rotation  | Auth   | [ ]    |
| 6  | T-Chain       | MPC wallet creation via FROST DKG                | Crypto | [ ]    |
| 7  | B-Chain       | Bridge lock/mint/burn/release full cycle         | Bridge | [ ]    |
| 8  | Z-Chain       | ZK proof generation and on-chain verification    | Crypto | [ ]    |
| 9  | Q-Chain       | Quantum-resistant signature schemes operational  | Crypto | [ ]    |
| 10 | A-Chain       | Agent execution, tool calls, state persistence   | AI     | [ ]    |
| 11 | G-Chain       | Governance proposals, voting, execution          | Gov    | [ ]    |
| 12 | O-Chain       | Oracle data feeds, attestation, freshness checks | Data   | [ ]    |
| 13 | R-Chain       | Research coordination, dataset registry          | Sci    | [ ]    |
| 14 | Beluga (L3)   | L3 subnet boots, settles to Zoo L2               | Core   | [ ]    |

## 2.2 GPU Acceleration Benchmarks

| # | Workload                             | Target                               | Owner  | Status |
|---|--------------------------------------|--------------------------------------|--------|--------|
| 1 | CEVM kernel execution (A100)         | $\geq 50\text{K tx/s}$               | Core   | [ ]    |
| 2 | TFHE gate evaluation (bootstrapping) | $\leq 12\text{ms per gate}$          | Crypto | [ ]    |
| 3 | BLS aggregate signature verify       | $\leq 2\text{ms} / 128 \text{ sigs}$ | Crypto | [ ]    |
| 4 | NTT for ZK proof generation          | $\leq 100\text{ms} / 2^{24}$         | Crypto | [ ]    |

## 2.3 Testing and Audit

| # | Item                                                | Owner  | Status |
|---|-----------------------------------------------------|--------|--------|
| 1 | E2E Playwright suite: all chains green              | QA     | [ ]    |
| 2 | Red team round 1 complete, findings remediated      | SecOps | [ ]    |
| 3 | Red team round 2 complete, findings remediated      | SecOps | [ ]    |
| 4 | Red team round 3 complete, 0 open findings          | SecOps | [ ]    |
| 5 | External audit report received and accepted         | SecOps | [ ]    |
| 6 | Formal proofs: 86 Lean 4 theorems compiled, 0 sorry | Formal | [ ]    |
| 7 | Bridge tested: LUX mainnet ↔ ZOO testnet            | Bridge | [ ]    |
| 8 | Fuzz testing: 72h continuous, 0 crashes             | QA     | [ ]    |

## 3 Rehearsal: Testnet to Staging

Rehearsal runs on `lux-test-k8s`. The full launch-day sequence is executed end-to-end. Any failure resets the rehearsal from step 1.

### 3.1 Genesis and Bootstrap

| # | Step                                                           | Owner  | Status |
|---|----------------------------------------------------------------|--------|--------|
| 1 | Generate genesis blocks for all 14 chains                      | Core   | [ ]    |
| 2 | Validator key ceremony: generate staking TLS certs via K-Chain | Auth   | [ ]    |
| 3 | P-Chain L1 creation transaction submitted and confirmed        | Core   | [ ]    |
| 4 | Chain creation transactions for all 14 VMs confirmed           | Core   | [ ]    |
| 5 | Validator registration (5 founding nodes) on P-Chain           | Core   | [ ]    |
| 6 | MPC wallet creation via T-Chain (FROST DKG for bridge keys)    | Crypto | [ ]    |
| 7 | DID issuance for all 5 validators via I-Chain                  | Auth   | [ ]    |
| 8 | Verify all 14 chain RPC endpoints responding                   | Core   | [ ]    |

### 3.2 Load and Chaos Testing

| # | Test                                                               | Owner  | Status |
|---|--------------------------------------------------------------------|--------|--------|
| 1 | Sustained throughput: 100K+ blocks produced across all chains      | Perf   | [ ]    |
| 2 | EVM sustained load: 10K tx/s for 1 hour, no degradation            | Perf   | [ ]    |
| 3 | Bridge stress: 1000 concurrent lock/mint operations                | Bridge | [ ]    |
| 4 | Kill 1 of 5 validators, verify chain progress continues            | Chaos  | [ ]    |
| 5 | Kill 2 of 5 validators, verify chain halts (BFT threshold)         | Chaos  | [ ]    |
| 6 | Restore killed validators, verify chain resumes within 30s         | Chaos  | [ ]    |
| 7 | Network partition: split 3/2, verify majority partition progresses | Chaos  | [ ]    |
| 8 | Disk full simulation: verify graceful degradation, no corruption   | Chaos  | [ ]    |

### 3.3 Rehearsal Sign-off

| # | Gate                                                              | Approver | Status |
|---|-------------------------------------------------------------------|----------|--------|
| 1 | All genesis/bootstrap steps completed without manual intervention | CTO      | [ ]    |
| 2 | Load test targets met                                             | CTO      | [ ]    |
| 3 | Chaos test recovery verified                                      | CTO      | [ ]    |
| 4 | Rehearsal log archived to IPFS                                    | DevOps   | [ ]    |

## 4 Launch Day

Launch proceeds only after rehearsal sign-off. All times are UTC. A launch coordinator runs the sequence; no ad-hoc changes permitted.

### 4.1 Genesis and Validators

| # | Action                                                  | Owner  | Status |
|---|---------------------------------------------------------|--------|--------|
| 1 | Genesis block signed by founding validators (multisig)  | Core   | [ ]    |
| 2 | Bootstrap 5 GPU validator nodes on <code>lux-k8s</code> | DevOps | [ ]    |
| 3 | P-Chain L1 creation confirmed on LUX mainnet            | Core   | [ ]    |
| 4 | All 14 chain creation transactions confirmed            | Core   | [ ]    |
| 5 | Validator registration confirmed for 5 founding nodes   | Core   | [ ]    |

### 4.2 Service Activation

| # | Action                                                          | Owner  | Status |
|---|-----------------------------------------------------------------|--------|--------|
| 1 | Verify all 14 chain RPCs responding via gateway                 | Core   | [ ]    |
| 2 | Bridge activation: LUX mainnet ↔ ZOO mainnet                    | Bridge | [ ]    |
| 3 | Bridge: first transfer confirmed both directions                | Bridge | [ ]    |
| 4 | Blockscout explorer live at <code>explore.zoo.ngo</code>        | DevOps | [ ]    |
| 5 | Explorer indexing all 14 chains                                 | DevOps | [ ]    |
| 6 | DNS cutover: <code>api.zoo.ngo</code> points to mainnet gateway | DevOps | [ ]    |
| 7 | DNS cutover: <code>explore.zoo.ngo</code> points to Blockscout  | DevOps | [ ]    |
| 8 | TLS certificates valid and proxied via Cloudflare               | DevOps | [ ]    |

### 4.3 Monitoring

| # | Action                                                   | Owner  | Status |
|---|----------------------------------------------------------|--------|--------|
| 1 | Prometheus scraping all validator and chain metrics      | DevOps | [ ]    |
| 2 | Grafana dashboards live: consensus, throughput, latency  | DevOps | [ ]    |
| 3 | Alerting rules active: missed blocks, high latency, disk | DevOps | [ ]    |
| 4 | PagerDuty escalation configured for critical alerts      | DevOps | [ ]    |
| 5 | Structured logging flowing to observability stack        | DevOps | [ ]    |

## 4.4 Launch Sign-off

| # | Gate                               | Approver | Status |
|---|------------------------------------|----------|--------|
| 1 | All 14 chains producing blocks     | CTO      | [ ]    |
| 2 | Bridge operational both directions | CTO      | [ ]    |
| 3 | Explorer indexing confirmed        | CTO      | [ ]    |
| 4 | Monitoring and alerting verified   | CTO      | [ ]    |
| 5 | Public announcement authorized     | CEO      | [ ]    |

## 5 Post-Launch

### 5.1 24-Hour Monitoring Window

| # | Item                                                   | Owner  | Status |
|---|--------------------------------------------------------|--------|--------|
| 1 | On-call rotation staffed for 24h (2 engineers minimum) | DevOps | [ ]    |
| 2 | Block production continuous, no gaps >5s               | Core   | [ ]    |
| 3 | Bridge transfers monitored, no stuck transactions      | Bridge | [ ]    |
| 4 | Memory and CPU within resource limits on all nodes     | DevOps | [ ]    |
| 5 | No consensus failures or forks detected                | Core   | [ ]    |

### 5.2 Validator Onboarding

| # | Item                                                       | Owner | Status |
|---|------------------------------------------------------------|-------|--------|
| 1 | Community validator documentation published                | Docs  | [ ]    |
| 2 | Validator onboarding guide tested by external party        | QA    | [ ]    |
| 3 | First community validator registered (target: launch +48h) | Core  | [ ]    |
| 4 | Gradual ramp: 10 validators by week 1, 25 by week 4        | Core  | [ ]    |

### 5.3 Governance and Staking

| # | Item                                                      | Owner    | Status |
|---|-----------------------------------------------------------|----------|--------|
| 1 | Staking portal activated on <code>app.zoo.ngo</code>      | Frontend | [ ]    |
| 2 | Staking contract verified on explorer                     | Core     | [ ]    |
| 3 | First governance vote: ZIP-001 submitted via G-Chain      | Gov      | [ ]    |
| 4 | ZIP-001 voting period completes, result executed on-chain | Gov      | [ ]    |

## 6 Rollback Plan

If mainnet exhibits critical failures (consensus divergence, bridge exploit, data corruption), the following rollback procedure activates.

### 6.1 Chain Halt Conditions

A chain halt is triggered if **any** of the following occur:

1. Consensus divergence: two or more validators on different forks for >60s.
2. Bridge exploit: unauthorized mint or release detected.

3. Data corruption: state root mismatch across >1 validator.
4. Critical vulnerability: zero-day with active exploitation.

## 6.2 Halt Procedure

| # | Action                                                          | Owner  |
|---|-----------------------------------------------------------------|--------|
| 1 | CTO authorizes chain halt                                       | CTO    |
| 2 | Disable bridge (freeze lock/mint contracts on both sides)       | Bridge |
| 3 | Coordinate validator shutdown via authenticated channel         | DevOps |
| 4 | All 5 founding validators stop block production                 | DevOps |
| 5 | Public status page updated: <code>status.zoo.ngo</code>         | Comms  |
| 6 | Post-mortem investigation begins (target: root cause within 4h) | Core   |

## 6.3 Genesis Reset (Last Resort)

If the chain state is unrecoverable:

1. Export last-known-good state snapshot from all validators.
2. Generate new genesis from snapshot (preserving balances and contract state).
3. Validators re-key via K-Chain key ceremony.
4. Re-execute bootstrap sequence (Section 3.1).
5. Bridge re-activation with fresh MPC keys via T-Chain.
6. Independent audit of new genesis state before restart.

This procedure destroys transaction history after the snapshot. It is used only when all other recovery options are exhausted.

## 6.4 Communication Plan

| # | Action                                                | Owner |
|---|-------------------------------------------------------|-------|
| 1 | Status page updated within 5 minutes of halt decision | Comms |
| 2 | Community notification via Discord, Twitter, blog     | Comms |
| 3 | Hourly updates until resolution                       | Comms |
| 4 | Post-mortem published within 72 hours                 | Core  |

## 7 Appendix: Chain Reference

| Chain         | VM ID                 | Purpose                                           |
|---------------|-----------------------|---------------------------------------------------|
| EVM (C-Chain) | <code>cevm</code>     | General-purpose smart contracts (GPU-accelerated) |
| DEX (D-Chain) | <code>dex</code>      | Native decentralized exchange with order book     |
| FHE (F-Chain) | <code>fhe</code>      | Fully homomorphic encryption compute              |
| I-Chain       | <code>identity</code> | Decentralized identity (W3C DID)                  |
| K-Chain       | <code>keychain</code> | Key management and certificate authority          |
| T-Chain       | <code>thresh</code>   | Threshold signatures and MPC wallets              |
| B-Chain       | <code>bridge</code>   | Cross-chain bridge operations                     |
| Z-Chain       | <code>zkvm</code>     | Zero-knowledge proof execution                    |
| Q-Chain       | <code>pqc</code>      | Post-quantum cryptographic operations             |
| A-Chain       | <code>agent</code>    | AI agent execution and orchestration              |
| G-Chain       | <code>gov</code>      | Governance proposals and voting                   |
| O-Chain       | <code>oracle</code>   | Decentralized oracle network                      |
| R-Chain       | <code>research</code> | Research coordination and datasets                |
| Beluga (L3)   | <code>beluga</code>   | High-throughput L3 subnet                         |